Online Appendix to
Optimal Dynamic Selection Under Costly Evaluation: Evidence from Graduate Admissions

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A Waitlist Models

In the matriculation-dynamic model in Section 4.2.1, the committee considers matriculation probabilities in Round 1. However, the degree to which it does so is small enough that the static model without matriculation serves as a static approximation for the matriculation-dynamic model. Committee members at the university featured in this paper also indicate that the committee does not consider matriculation probabilities in Round 1. The reason is that this university is ranked highly enough that it can fill its cohort with applicants who have strong objective and subjective variables, leaving Round 2 with enough slack to be selective on success regardless of Round 1's matriculation weighting. However, this university does consider matriculation probabilities after Round 1, where additional variables, such as *Letters Terms (%) PC1*, predict matriculation.[1]

Therefore, in this appendix, I introduce the waitlist and expand the cutoff rule framework from two rounds to three. Success probabilities equal, the committee favors likely matriculants in Round 2 to avoid losing high-quality applicants who would have matriculated by suboptimally keeping those applicants waiting until Round 3 for an acceptance. I also make the simplifying assumption that matriculation probabilities do not enter the Round 1 objective.

A.1 Asymmetric Waitlist Models

An admissions committee does not send out all its acceptances at once, but rather in batches. It starts by accepting its most preferred applicants, and then it puts the others who survived Round 1 on a waitlist. Then provided not all its early acceptances matriculate, it accepts applicants off the waitlist and ultimately moves down the waitlist until it has filled its cohort. I model this process as concisely as possible by increasing the number of rounds to three. As shown in Figure 1, the committee observes only the objective variables in Round 1 and either transitions the applicant to Round 2 or rejects them out of courtesy. But in Round 2, instead of accepting or rejecting the survivors based on the objective and subjective variables, it accepts or waitlists them.

Then in Round 3, it accepts or rejects the waitlisted applicants. Between Rounds 2 and 3, it observes how many of the early acceptances matriculate and makes its Round 3 decisions accordingly. If more early acceptances matriculate, it accepts fewer applicants off the waitlist because there are fewer openings left to fill. Intuitively, this situation is better than the other way around. More early acceptances matriculating means the committee gets more of its top choices. In contrast, if the committee has to accept many applicants off the waitlist, it may have to lower its standards because applicants it could have gotten with early offers may have already committed to other schools. Therefore, it should err on the side of giving Round 2 offers to applicants who are relatively likely to matriculate. That said, it would be incorrect to have the committee value

[1] Per Table 2b of Section 3, the matriculation-predicting variables in x are *Winsorized Age* (-), *U.S.* (+), *International Asian* (+), *GRE Quantitative Percentile* (-), and *Work Experience* (+). The matriculation-predicting variables in z are *Advanced Math* (-), *Letters Terms (%) PC1* (+), *CV Coding* (-), and *Essay Economics Terms (%)* (+).

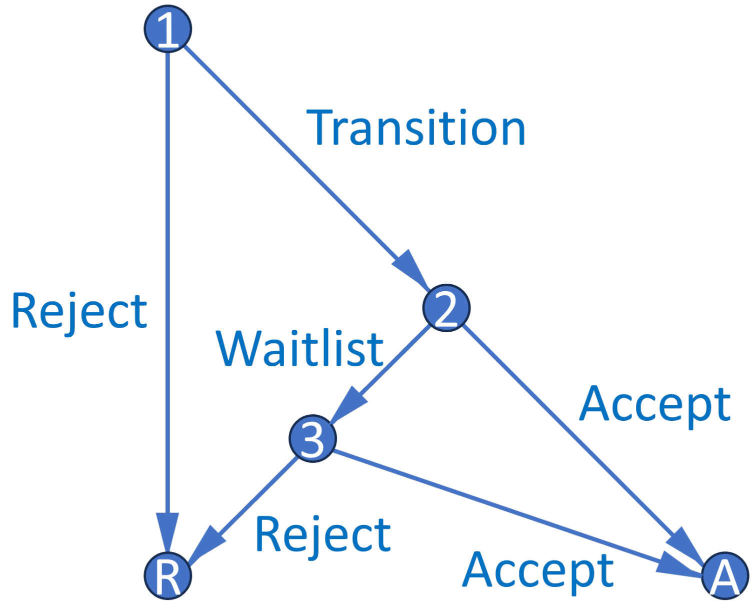


Figure 1: Waitlist Decision Graph

applicants by a combination of their success and matriculation probabilities, as that would be the cutoff rule solution to a portfolio choice problem with a constraint on the number of acceptances, not matriculants. The matriculation probabilities must instead appear elsewhere in the objective.

As such, I set up the model by letting the Round 3 cutoff probability be $P(y_3^* = S|\overline{m_{2A}})$ instead of $P(y_3^* = S|t)$. Here, $\overline{m_{2A}}$ is the average early accepted applicant's matriculation probability in a given year, which measures how likely the the applicant is to matriculate if accepted. $P(\text{Cutoff})_3$ increases in $\overline{m_{2A}}$ because a larger $\overline{m_{2A}}$ means there will be fewer openings for waitlisted applicants. Given that, what information available at the start of Round 2 determines $\overline{m_{2A}}$? For all applicants in each year, I define $\ddot{m}_2$ as $\overline{m_2}$ but with m_2 excluded, where m_2 is the applicant's own matriculation probability. Within each year, a low m_2 necessarily implies a high $\ddot{m}_2$, since $\overline{m_2}$ is a within-year average. I then estimate $f_{\overline{m_{2A}}|\ddot{m}_2}$ on all transitioned applicants; the parameter estimate should be positive provided there exists variation in $\overline{m_2}$ across years. For example, by decreasing acceptance probabilities, Covid may have made applicants more likely to matriculate if accepted. I assume that the effect of the average matriculation probability of the applicants the committee can accept ($\ddot{m}_2$) on that of the applicants it accepts ($\overline{m_{2A}}$) is stationary across years, and therefore can be used to forecast changes in $\overline{m_{2A}}$ based on changes in $\ddot{m}_2$ resulting from changes in d_2.

The model implies that the optimal dynamic decision in Round 2 may not be the same as the optimal myopic one. Specifically, a committee may waitlist, as opposed to accept, an applicant with a low m_2 even if their success probability is greater than an applicant with a higher m_2. The reason is that compared with a high m_2 applicant, a low m_2 applicant has a relatively high $\ddot{m}_2$. That implies a high $\overline{F}_{\overline{m_{2A}}|\ddot{m}_2}$, which in turn implies a high Round 3 cutoff probability, Round 3 value function, and payoff for waitlisting the low m_2 applicant in Round 2. Suppose that a hypothetical committee does not think dynamically; that is, removing these memory parameters increases the AIC. However, if $\ddot{m}_2$ affects $\overline{F}_{\overline{m_{2A}}|\ddot{m}_2}$ and in turn $\mathbb{E}[P(\text{Cutoff})_3]$, perhaps the committee should.

That said, why is the committee not able to accept the low m_2 applicant in Round 2 and then, if this applicant does not matriculate, fill the additional opening with the high m_2 applicant that it necessarily waitlisted? There are two reasons why such a strategy may not pay off. First, all else equal, matriculation probabilities are expected to decline between Rounds 2 and 3 because applicants are more likely to go where they feel wanted; whether and by how much they decline is an empirical matter. If matriculation probabilities were to decline uniformly for likely vs. unlikely matriculants, the committee could still get away with accepting unlikely matriculants and filling any extra openings with likely matriculants. However, if applicants with high matriculation probabilities have more room for those matriculation probabilities to fall when the committee keeps them waiting until Round 3, then the better decision is to accept them in Round 2.

Second, matriculation probabilities can change unpredictably between Rounds 2 and 3 because applicants receive acceptances, waitlists, and rejections from multiple universities, communicate with faculty and graduate students at universities where they were accepted or waitlisted, and attend open houses. These events, which are unknown to the committee at the start of Round 2, make it so that waitlisting a likely matriculant under the belief that they will still matriculate if waitlisted is a risky endeavor; by Round 3, the committee may find out too late that this applicant is now unlikely to matriculate. Therefore, it is safer to accept this applicant in Round 2.[2]

As such, it is clear that accepting applicants who want to matriculate benefits the program. I also discount the expected Round 3 value with $\beta \in (0,1)$ so that the payoff from accepting an applicant in Round 3 is less than that from accepting them in Round 2. The discount factor captures the committee's perceived cost of deferring the final decision to Round 3 and can be interpreted as the chance of losing a representative applicant to another university; loosely, it reflects the ratio of expected matriculation probabilities for Round 3 vs. Round 2 acceptances. Of course, without a capacity constraint, the committee could give early acceptances to the low and high m_2 applicants, but because of the constraint, it must avoid overshooting its target cohort size. I now present the value functions for the waitlist-dynamic model, starting with Round 3:

$$v_3(x,z,t,\overline{k},\overline{m_{2A}},\varepsilon_3) = \max_{d_3 \in \{A,R\}} u_3(x,z,t,\overline{k},\overline{m_{2A}},d_3) + \varepsilon_3(d_3), \text{ such that:} \tag{1}$$

$$u_3(x,z,t,\overline{k},\overline{m_{2A}},d_3) = P(y = S|x,z,t,\overline{k})\mathbb{1}(d_3 = A) + P(y_3^* = S|\overline{m_{2A}})\mathbb{1}(d_3 = R).$$

The Round 2 value function is:

$$v_2(x,z,t,\overline{k},\ddot{m}_2,\varepsilon_2) = \max_{d_2 \in \{A,W\}} u_2(x,z,t,\overline{k},\ddot{m}_2,d_2) + \varepsilon_2(d_2), \text{ such that:} \tag{2}$$

[2] In Appendix A.3, I present a more general version of the model in which the Round 3 cutoff probability is a function of $\overline{m_{2A}}$ and $\overline{m_3}$, which is the average matriculation probability of the waitlisted applicants and depends on m_3, the *ex-ante* probability of matriculating if accepted in Round 3. In this model, the effect of matriculation probabilities on Round 2 acceptance probabilities is theoretically ambiguous and depends on the parameter values.

$u_2(x,z,t,\bar{k},\ddot{m}_2,d_2) = P(y=S|x,z,t,\bar{k})\mathbb{1}(d_2=A) + \beta\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m_{2A}},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2]\mathbb{1}(d_2=W)$, such that:

$$\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m_{2A}},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2] = \int_{\widetilde{\overline{m_{2A}}}} \sigma \log\left(\sum_{\tilde{d}_3\in\{A,R\}} e^{u_3(x,z,t,\bar{k},\widetilde{\overline{m_{2A}}},\tilde{d}_3)/\sigma}\right) dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2).$$

The Round 1 value function depends on whether I include a conditional expectation, as in the dynamic model, or whether I do not, as in the static model. I refer to the three-round model without a Round 1 conditional expectation as the waitlist-hybrid model, since it has a static Round 1 and dynamic Round 2. The waitlist-dynamic model's Round 1 value function is:

$$v_1(x,t,\bar{k},\varepsilon_1) = \max_{d_1\in\{T,R\}} u_1(x,t,\bar{k},d_1) + \varepsilon_1(d_1), \text{ such that:} \tag{3}$$

$$u_1(x,t,\bar{k},d_1) = \mathbb{E}[v_2(x,z,t,\bar{k},\ddot{m}_2,\varepsilon_2)|x,t,\bar{k}]\mathbb{1}(d_1=T) + L(y_1^*=S|t)\mathbb{1}(d_1=R), \text{ such that:}$$

$$\mathbb{E}[v_2(x,z,t,\bar{k},\ddot{m}_2,\varepsilon_2)|x,t,\bar{k}] = \int_{\tilde{z}} \sigma \log\left(\sum_{\tilde{d}_2\in\{A,W\}} e^{u_2(x,\tilde{z},t,\bar{k},\ddot{m}_2(x,\tilde{z},t),\tilde{d}_2)/\sigma}\right) dF_{z|x,t}(\tilde{z}|x,t).$$

In contrast, the utility function in the waitlist-hybrid's Round 1 value function is:

$$u_1(x,t,\bar{k},d_1) = P(y=S|x,t,\bar{k})\mathbb{1}(d_1=T) + P(y_1^*=S|t)\mathbb{1}(d_1=R).$$

Expressed as simply as possible, the restricted CCPs are:

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states},d_r)/\sigma)}{\sum_{\tilde{d}_r\in D_r}\exp(u_r(\text{states},\tilde{d}_r)/\sigma)}. \tag{4}$$

The waitlist-dynamic model's unrestricted CCPs, success probabilities, and cutoff probabilities are $P_{d_1} = P\big(d_1=T\big|x,t;\theta_T\big)$, $P_{d_2} = P\big(d_2=A\big|x,z,\ddot{m}_2;\theta_{A_2}\big)$, $P_{d_3} = P\big(d_3=A\big|x,z,\overline{m_{2A}};\theta_{A_3}\big)$, $P_y = P\big(y=S\big|x,z,t,\bar{k};\theta_S\big)$, $L_{y_1^*} = L\big(y_1^*=S\big|t;\theta_{S_1^*}\big)$, and $P_{y_3^*} = P\big(y_3^*=S\big|\overline{m_{2A}};\theta_{S_3^*}\big)$. These probabilities have the same specifications as in the dynamic model (see Equations (11)–(13) for the functional forms).[3] Let applicants 1 to N_W be waitlisted, N_W+1 to N_T accepted in Round 2, and N_T+1 to N rejected in Round 1. Then the unrestricted waitlist-dynamic likelihood is:

$$\begin{aligned}\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} f\big(z_n^c|z_n^{\neg c},x_n,t_n;\theta_{F_z}\big) f\big(\overline{m_{2A,n}}|\ddot{m}_{2,n};\theta_{F_m}\big)\\ P_{d_{1,n}}(\theta_T)P_{d_{2,n}}(\theta_{A_2})^{\mathbb{1}(d_{2,n}=A)}(1-P_{d_{2,n}}(\theta_{A_2}))^{\mathbb{1}(d_{2,n}=W)}\prod_{n=1}^{N_W}P_{d_{3,n}}(\theta_{A_3})^{\mathbb{1}(d_{3,n}=A)}(1-P_{d_{3,n}}(\theta_{A_3}))^{\mathbb{1}(d_{3,n}=R)}\\ \prod_{n=N_T+1}^{N}\big(1-P_{d_{1,n}}(\theta_T)\big)\prod_{n=1}^{N} f\big(z_n^{\neg c}|x_n,t_n;\theta_{F_z}\big)P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}.\end{aligned} \tag{5}$$

[3] $L_{y_1^*}$ is parameterized as exponential because it can exceed 1. However, in the waitlist-hybrid model, it is denoted as $P_{y_1^*}$ and parameterized as logistic. The waitlist-hybrid model also includes $P_{y_1}\big(y=S\big|x,t,\bar{k};\theta_{S_1}\big)$.

As with the dynamic and matriculation-dynamic likelihoods, a Heckman correction is necessary for $f_{z|x,t}$. I also estimate the parameters of $m_2 = P_{m_2}(\theta_{M_2})$, and therefore $\ddot{m}_2$, in a first-stage likelihood where I restrict the sample to Round 2 acceptances. The dependent variable equals 1 if the applicant matriculates and 0 otherwise, and the independent variables are x, z, and t. Because $\ddot{m}_2$ is a generated regressor in both the unrestricted and restricted likelihoods, I estimate standard errors per Murphy and Topel (1985) and Hardin (2002). The first-stage likelihood is:

$$\mathcal{L}^{M_2}_{N_T-N_W}(\theta_{M_2}) = \prod_{n=N_W+1}^{N_T} P_{m_{2,n}}(\theta_{M_2})^{\mathbb{1}(m_{2,n}=M)}(1-P_{m_{2,n}}(\theta_{M_2}))^{\mathbb{1}(m_{2,n}=D)}. \tag{6}$$

To show that the waitlist models' dynamics work as intended, I prove the following theorem:

Theorem 1. *Success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.*

Proof. Suppose there are two arbitrary applicants in year t indexed by H and L, such that $P(y=S|x^H,z^H,t,\bar{k}) = P(y=S|x^L,z^L,t,\bar{k})$, but $m_2^H > m_2^L$. That is, the applicants have equal success probabilities, but H has the higher matriculation probability. Since $\ddot{m}_2$ is the average matriculation probability of all applicants in t with m_2 excluded, $\ddot{m}_2^H < \ddot{m}_2^L$ by construction. Given this inequality, I will show $P(d_2=A|x^H,z^H,t,\bar{k},\ddot{m}_2^H) > P(d_2=A|x^L,z^L,t,\bar{k},\ddot{m}_2^L)$. It is the case that:

$$P(d_2=A|x,z,t,\bar{k},\ddot{m}_2) = \frac{\exp(P(y=S|x,z,t,\bar{k})/\sigma)}{\exp(P(y=S|x,z,t,\bar{k})/\sigma)+\exp(\beta\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m}_{2A},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2]/\sigma)}, \text{ where}$$

$$\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m}_{2A},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2] = \int_{\widetilde{\overline{m}_{2A}}} \sigma\log\left(e^{P(y=S|x,z,t,\bar{k})/\sigma}+e^{P(y_3^*=S|\widetilde{\overline{m}_{2A}})/\sigma}\right)dF_{\overline{m}_{2A}|\ddot{m}_2}(\widetilde{\overline{m}_{2A}}|\ddot{m}_2). \tag{7}$$

The only place $\ddot{m}_2$ appears in $P(d_2=A|x,z,t,\bar{k},\ddot{m}_2)$ is in the distribution of $\overline{m}_{2A}$ conditional on $\ddot{m}_2$. I parameterize $\overline{m}_{2A}|\ddot{m}_2 \sim \mathcal{N}(\ddot{m}_2\beta_{\ddot{m}_2},\sigma^2_{\ddot{m}_2})$, where by assumption $\beta_{\ddot{m}_2} > 0$. $\beta_{\ddot{m}_2}$ is positive because years when $\ddot{m}_2$ is high are years when matriculation probabilities are high in general. Using inter-year variation, this relationship results in a greater expected $\overline{m}_{2A}$ in those years. Then:

$$F_{\overline{m}_{2A}|\ddot{m}_2}(\overline{m}_{2A}|\ddot{m}_2^H;\beta_{\ddot{m}_2},\sigma^2_{\ddot{m}_2}) = \Phi\left(\frac{\overline{m}_{2A}-\ddot{m}_2^H\beta_{\ddot{m}_2}}{\sigma_{\ddot{m}_2}}\right) > \Phi\left(\frac{\overline{m}_{2A}-\ddot{m}_2^L\beta_{\ddot{m}_2}}{\sigma_{\ddot{m}_2}}\right) = F_{\overline{m}_{2A}|\ddot{m}_2}(\overline{m}_{2A}|\ddot{m}_2^L;\beta_{\ddot{m}_2},\sigma^2_{\ddot{m}_2}). \tag{8}$$

Therefore, $\overline{m}_{2A}|\ddot{m}_2^L$ first-order stochastically dominates $\overline{m}_{2A}|\ddot{m}_2^H$. To apply this stochastic dominance, I must show that $\sigma\log\left(e^{P(y=S|x,z,t,\bar{k})/\sigma}+e^{P(y_3^*=S|\overline{m}_{2A})/\sigma}\right)$ is strictly increasing in $\overline{m}_{2A}$. But the only place $\overline{m}_{2A}$ appears in this expression is in $P(y_3^*=S|\overline{m}_{2A};\theta_{S_3^*}) = \frac{\exp\left(f_{S_3^*}(\overline{m}_{2A})\theta_{S_3^*}\right)}{1+\exp\left(f_{S_3^*}(\overline{m}_{2A})\theta_{S_3^*}\right)}$, where by assumption $f_{S_3^*}(\cdot)$ is strictly increasing and $\theta_{S_3^*} > 0$. $\theta_{S_3^*}$ is positive because the expected Round 3 cutoff probability is higher when early acceptees are more likely on average to matriculate. As such, I have that $P(y_3^*=S|\overline{m}_{2A})$, and in turn $\sigma\log\left(e^{P(y=S|x,z,t,\bar{k})/\sigma}+e^{P(y_3^*=S|\overline{m}_{2A})/\sigma}\right)$, are strictly increasing in $\overline{m}_{2A}$. Combining this result with $(\overline{m}_{2A}|\ddot{m}_2^L) \succ_{\text{FOSD}} (\overline{m}_{2A}|\ddot{m}_2^H)$, I obtain:

$$\begin{aligned}
\mathbb{E}[v_3(x^L, z^L, t, \overline{k}, \overline{m}_{2A}, \varepsilon_3)|x^L, z^L, t, \overline{k}, \ddot{m}_2^L] &= \int_{\widetilde{\overline{m}_{2A}}} \sigma \log\left(e^{P(y=S|x^L,z^L,t,\overline{k})/\sigma} + e^{P(y_3^*=S|\widetilde{\overline{m}_{2A}})/\sigma}\right) dF_{\overline{m}_{2A}|\ddot{m}_2}(\widetilde{\overline{m}_{2A}}|\ddot{m}_2^L) \\
&= \int_{\widetilde{\overline{m}_{2A}}} \sigma \log\left(e^{P(y=S|x^H,z^H,t,\overline{k})/\sigma} + e^{P(y_3^*=S|\widetilde{\overline{m}_{2A}})/\sigma}\right) dF_{\overline{m}_{2A}|\ddot{m}_2}(\widetilde{\overline{m}_{2A}}|\ddot{m}_2^L) \\
&> \int_{\widetilde{\overline{m}_{2A}}} \sigma \log\left(e^{P(y=S|x^H,z^H,t,\overline{k})/\sigma} + e^{P(y_3^*=S|\widetilde{\overline{m}_{2A}})/\sigma}\right) dF_{\overline{m}_{2A}|\ddot{m}_2}(\widetilde{\overline{m}_{2A}}|\ddot{m}_2^H) \\
&= \mathbb{E}[v_3(x^H, z^H, t, \overline{k}, \overline{m}_{2A}, \varepsilon_3)|x^H, z^H, t, \overline{k}, \ddot{m}_2^H]. \qquad (9)
\end{aligned}$$

The second equality holds since $P(y = S|x^H, z^H, t, \overline{k}) = P(y = S|x^L, z^L, t, \overline{k})$. Therefore, by Equation (7) and Inequality (9), I have that $P(d_2 = A|x^H, z^H, t, \overline{k}, \ddot{m}_2^H) > P(d_2 = A|x^L, z^L, t, \overline{k}, \ddot{m}_2^L)$.

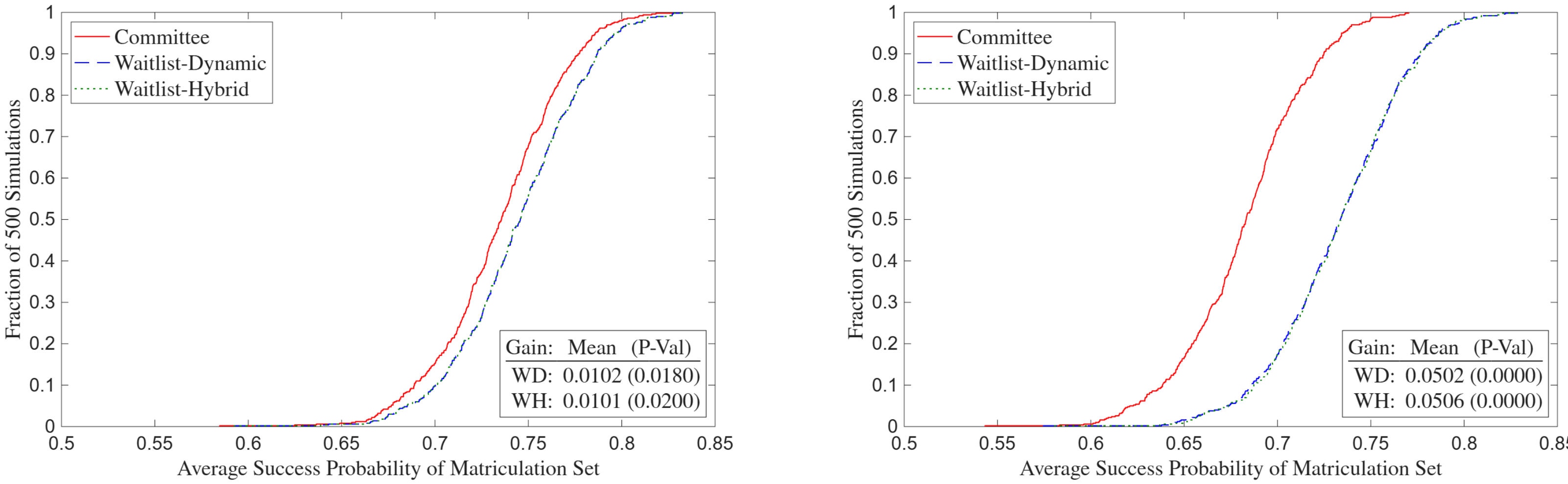
A.2 Asymmetric Waitlist Results

Unlike the two-round models, I do not provide estimation results for the waitlist models on actual data. The reason is that the relationship between $\overline{m_{2A}}$ and $\ddot{m}_2$ is only identified by variation across years, given that $\overline{m_{2A}}$ is constant within years. I have only five years of data, and the Covid shock, which may provide variation in $\overline{m_2}$ across years, is not part of the sample. As such, I have insufficient statistical power to estimate the F$\overline{m_{2A}}$ block of the likelihood, at least not until a potential future date when applicants from multiple application years after 2019 have completed their PhDs. Therefore, I stick to estimating the waitlist models on the same optimal and suboptimal simulated datasets that I use to demonstrate that the deviation gains test for the two-round models rejects optimality only when it is supposed to. In the simulated data, there are only two years, but the data are generated such that the relationship between $\overline{m_{2A}}$ and $\ddot{m}_2$ is identified.[4] The purpose of this simulated data estimation is to demonstrate that there exists a plausible data-generating process in which the waitlist models' parameters are identified.

The unrestricted waitlist-dynamic and hybrid estimates are in Tables 1 and 3. As expected, the $\ddot{m}_2$ parameter in the F$\overline{m_{2A}}$ block is positive and significant at the 1% level in both datasets. In addition, the $\ddot{m}_2$ parameter in the *Accept 2* block is negative in both datasets, which is also as expected. The reason is that the committee is more likely to give an early offer to an applicant with a high m_2 and therefore a low $\ddot{m}_2$. Finally, the $\overline{m_{2A}}$ parameter in the *Accept 3* block is negative and significant at the 1% level because a high $\overline{m_{2A}}$ means fewer spots will be open in Round 3 due to more Round 2 matriculants, and so it will be harder to get accepted off the waitlist.

Tables 2 and 4 contain restricted estimates for the waitlist-dynamic and hybrid models. In the waitlist-hybrid model, as with the two-round static model, the Round 1 cutoff probability is higher than the Round 3 cutoff probability in the year that is not *Year Transition/Matriculate More*.

[4] I set the weight that the committee puts on matriculation in Round 2 to be the same in both years, which keeps $\overline{m_{2A}}$ and $\ddot{m}_2$ strongly correlated. In the actual data, there is some variation in that weight across years, which along with only five years of data makes the correlation between $\overline{m_{2A}}$ and $\ddot{m}_2$ insignificantly different from zero.



(a) Optimal Committee

(b) Suboptimal Committee

Figure 2: Waitlist Deviation Gains

As such, the marginal applicant rejected in Round 1's $P(y=S|x,t,\bar{k})$ is greater than than the marginal applicant accepted into the program's $P(y=S|x,z,t,\bar{k})$, which may be undesirable for the committee. Moreover, in *Year Transition/Matriculate More*, the Round 3 cutoff probability is higher than the Round 1 cutoff probability not only because there are more Round 1 survivors, but also because those applicants have higher matriculation probabilities on average.

Finally, I run the deviation gains test for the waitlist models. For the waitlist-dynamic model, I rank the applicants in each year by $\mathbb{E}[v_2(x,z,t,\bar{k},\ddot{m}_2,\varepsilon_2)|x,t,\bar{k}] = \int_{\tilde{z}} \sigma \log\Big(e^{P(y=S|x,\tilde{z},t,\bar{k})/\sigma} + e^{\beta\mathbb{E}[v_3(x,\tilde{z},t,\bar{k},\widetilde{\overline{m_{2A}}},\tilde{\varepsilon}_3)|x,\tilde{z},t,\bar{k},\ddot{m}_2(x,\tilde{z},t)]/\sigma}\Big) dF_{z|x,t}(\tilde{z}|x,t)$ and transition the number of transitioned applicants in the data. Next, I rank the surviving applicants by the payoff difference $P(y=S|x,z,t,\bar{k}) - \beta\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m_{2A}},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2] = P(y=S|x,z,t,\bar{k}) - \beta\int_{\widetilde{\overline{m_{2A}}}} \sigma\log\Big(e^{P(y=S|x,z,t,\bar{k})/\sigma} + e^{P(y_3^*=S|\widetilde{\overline{m_{2A}}})/\sigma}\Big) dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m}}$ accept the number of early acceptances in the data, and waitlist the rest. Each early acceptance matriculates with early matriculation probability $P(m_2=M|x,z,t)$. After that, I rank the waitlisted applicants by $P(y=S|x,z,t,\bar{k})$ and accept them in descending order, each matriculating with late matriculation probability $P(m_3=M|x,z,t)$, until the number of early plus late matriculants equals that in the data. Lastly, I compute the average success probability of the matriculation set and compare it to the committee's set. For the waitlist-hybrid model, I instead rank the applicants in Round 1 by $P(y=S|x,t,\bar{k})$. The results are in Figure 2. Given the difficulty of creating a dataset that does not reject optimality at the 5% level for both the two-round and waitlist models, the optimal dataset's p-values are 0.0180 and 0.0200 for the waitlist-dynamic and hybrid models, respectively, albeit with a deviation gain of just 1 percentage point. Meanwhile, the suboptimal dataset's p-values are less than 0.0002, and its deviation gains are 5 percentage points.

Table 1: Simulated Unrestricted Waitlist-Dynamic Estimates

(a) Optimal Committee

	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success 1		Success 3		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-0.0249	-11.5885***	-2.2558***	-11.9709***	-1.6176***	-13.2754***	-1.6998***	-8.1374***	-1.1678***	-9.4807***	-1.3248***	-5.9143	-0.6836
	(0.0516)	(1.0232)	(0.1454)	(2.5515)	(0.3154)	(2.4119)	(0.2845)	(1.1381)	(0.1344)	(1.3435)	(0.1562)	(6.4966)	(0.7703)
GRE Quantitative Percentile		0.1236***	0.0241***	0.1097***	0.0148***	0.1387***	0.0178***	0.1079***	0.0155***	0.1150***	0.0161***	0.0842	0.0097
		(0.0118)	(0.0018)	(0.0271)	(0.0034)	(0.0265)	(0.0032)	(0.0133)	(0.0015)	(0.0145)	(0.0016)	(0.0645)	(0.0075)
Gender		0.2266	0.0441	0.0234	0.0032	0.3040	0.0389	0.1399	0.0201	0.2078	0.0290	0.4647	0.0537
		(0.1464)	(0.0283)	(0.2899)	(0.0392)	(0.3288)	(0.0421)	(0.1739)	(0.0249)	(0.1766)	(0.0246)	(0.9007)	(0.1052)
Demographic 1		-0.3014	-0.0587	0.5534	0.0748	-0.8832*	-0.1131*	-0.2533	-0.0364	-0.3380	-0.0472	-0.2627	-0.0304
		(0.1988)	(0.0386)	(0.4942)	(0.0665)	(0.4623)	(0.0578)	(0.2359)	(0.0338)	(0.2456)	(0.0342)	(0.7742)	(0.0879)
Demographic 2		0.1538	0.0299	0.5137	0.0694	-0.0523	-0.0067	-0.1179	-0.0169	-0.1354	-0.0189	-0.3955	-0.0457
		(0.2215)	(0.0431)	(0.5205)	(0.0700)	(0.4528)	(0.0580)	(0.2581)	(0.0371)	(0.2652)	(0.0371)	(1.2095)	(0.1391)
Advanced Math				0.5067*	0.0685*	0.8622***	0.1104***			0.6827***	0.0954***	1.1444	0.1323
				(0.2907)	(0.0390)	(0.3306)	(0.0421)			(0.1770)	(0.0237)	(0.9852)	(0.1080)
Committee Score				0.1061	0.0143	0.0378	0.0048			0.0467	0.0065	-0.9116**	-0.1054***
				(0.0689)	(0.0093)	(0.0672)	(0.0086)			(0.0672)	(0.0094)	(0.3783)	(0.0317)
Missing Committee Score										0.2177	0.0304		
										(0.1896)	(0.0264)		
Year Transition/Matriculate More		0.9840***	0.1915***					-0.0587	-0.0084	-0.0466	-0.0065	3.3941***	0.3923***
		(0.1472)	(0.0263)					(0.1690)	(0.0242)	(0.1740)	(0.0243)	(0.7689)	(0.0698)
Program Rank								-0.0310***	-0.0045***	-0.0309***	-0.0043***		
								(0.0019)	(0.0002)	(0.0019)	(0.0002)		
$\ddot{m}_2$	1.0896***			-1.5992**	-0.2161**								
	(0.0851)			(0.7875)	(0.1049)								
$\overline{m_{2A}}$						-2.0178***	-0.2584***						
						(0.7700)	(0.0964)						
$\sqrt{\text{MSE}}$	0.0015***												
	(0.0005)												
Obs | Param | LL | AIC	1000	42	191.4	-298.8									

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success 1		Success 3		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-0.0496*	-11.2295***	-2.1801***	-6.7113***	-1.0637***	-7.1639***	-0.8002***	-8.2825***	-1.1913***	-9.7587***	-1.3686***	-1.5054	-0.2179
	(0.0259)	(1.0144)	(0.1474)	(2.0007)	(0.3043)	(2.5070)	(0.2796)	(1.1392)	(0.1341)	(1.3345)	(0.1538)	(3.8095)	(0.5534)
GRE Quantitative Percentile		0.1226***	0.0238***	0.0387*	0.0061*	0.0717***	0.0080**	0.1097***	0.0158***	0.1175***	0.0165***	0.0362	0.0052
		(0.0117)	(0.0018)	(0.0215)	(0.0034)	(0.0278)	(0.0031)	(0.0133)	(0.0015)	(0.0145)	(0.0016)	(0.0435)	(0.0063)
Gender		0.2575*	0.0500*	-0.2767	-0.0439	0.4102	0.0458	0.1355	0.0195	0.1984	0.0278	-0.3519	-0.0509
		(0.1469)	(0.0283)	(0.2746)	(0.0431)	(0.3417)	(0.0380)	(0.1741)	(0.0250)	(0.1766)	(0.0247)	(0.5435)	(0.0788)
Demographic 1		-0.8293***	-0.1610***	0.2577	0.0408	-1.1548**	-0.1290**	-0.2776	-0.0399	-0.3897	-0.0547	0.3291	0.0476
		(0.1966)	(0.0371)	(0.3774)	(0.0594)	(0.5110)	(0.0563)	(0.2344)	(0.0337)	(0.2462)	(0.0343)	(0.7092)	(0.1031)
Demographic 2		0.0797	0.0155	0.5369	0.0851	-0.3696	-0.0413	-0.1809	-0.0260	-0.2068	-0.0290	-0.7626	-0.1104
		(0.2135)	(0.0414)	(0.3886)	(0.0605)	(0.4652)	(0.0521)	(0.2566)	(0.0369)	(0.2636)	(0.0370)	(0.7932)	(0.1090)
Advanced Math				0.2363	0.0375	0.4866	0.0544			0.6387***	0.0896***	0.3977	0.0576
				(0.2639)	(0.0417)	(0.3566)	(0.0399)			(0.1763)	(0.0238)	(0.6444)	(0.0922)
Committee Score				0.3737***	0.0592***	0.0653	0.0073			0.0626	0.0088	-0.6714***	-0.0972***
				(0.0674)	(0.0096)	(0.0837)	(0.0093)			(0.0657)	(0.0092)	(0.1978)	(0.0217)
Missing Committee Score										0.2535	0.0356		
										(0.1896)	(0.0264)		
Year Transition/Matriculate More		1.1108***	0.2156***					-0.0767	-0.0110	-0.0473	-0.0066	2.9420***	0.4259***
		(0.1479)	(0.0257)					(0.1688)	(0.0243)	(0.1757)	(0.0247)	(0.6724)	(0.0579)
Program Rank								-0.0304***	-0.0044***	-0.0303***	-0.0043***		
								(0.0019)	(0.0002)	(0.0019)	(0.0002)		
$\ddot{m}_2$	0.8882***			-0.5496	-0.0871								
	(0.0488)			(0.5706)	(0.0900)								
$\overline{m_{2A}}$						-2.9219***	-0.3264***						
						(0.9936)	(0.1102)						
$\sqrt{\text{MSE}}$	0.0010***												
	(0.0002)												
Obs | Param | LL | AIC	1000	42	374.8	-665.5									

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 2: Simulated Restricted Waitlist-Dynamic Estimates

(a) Optimal Committee

	$\overline{m_{2A}}$	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Year Transition/Matriculate Fewer | Intercept	-0.0249	0.6925***	66.65%	0.4942	70.64%	-8.0628***	-9.5820***	-5.9143		
	(0.0516)	(0.1818)	(58.33%, 74.05%)	(0.3513)	(57.55%, 81.02%)	(1.1622)	(1.3341)	(6.4966)		
GRE Quantitative Percentile						0.1062***	0.1154***	0.0842		
						(0.0134)	(0.0146)	(0.0645)		
Gender						0.1999**	0.1696	0.4647		
						(0.1018)	(0.1358)	(0.9007)		
Demographic 1						-0.2482*	-0.3098	-0.2627		
						(0.1353)	(0.1912)	(0.7742)		
Demographic 2						0.0407	-0.0318	-0.3955		
						(0.1511)	(0.1962)	(1.2095)		
Advanced Math							0.6616***	1.1444		
							(0.1424)	(0.9852)		
Committee Score							0.0560	-0.9117**		
							(0.0425)	(0.3783)		
Missing Committee Score							0.2167			
							(0.1838)			
Year Transition/Matriculate More		-0.8491***	46.09%		88.08%	-0.0604	-0.0289	3.3941***		
		(0.2179)	(39.64%, 52.68%)		(65.79%, 96.60%)	(0.1667)	(0.1727)	(0.7689)		
Program Rank						-0.0310***	-0.0310***			
						(0.0019)	(0.0019)			
$\tilde{m}_2$	1.0896***									
	(0.0851)									
$\overline{m_{2A}}$				2.3665*						
				(1.2823)						
$\sqrt{\text{MSE}}$	0.0015***									
	(0.0005)									
Beta									0.9490***	
									(0.0391)	
Sigma										0.1926***
										(0.0246)
Obs | Param | LL | AIC	1000	26	186.8	-321.6						

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	$\overline{m_{2A}}$	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Year Transition/Matriculate Fewer | Intercept	-0.0496*	0.8342***	69.72%	0.8216	80.71%	-8.6331***	-8.3628***	-1.5054		
	(0.0259)	(0.2171)	(60.07%, 77.90%)	(0.7603)	(61.39%, 91.67%)	(1.1831)	(1.4735)	(3.8095)		
GRE Quantitative Percentile						0.1147***	0.0947***	0.0362		
						(0.0137)	(0.0156)	(0.0435)		
Gender						0.2301**	0.1269	-0.3519		
						(0.1099)	(0.1412)	(0.5435)		
Demographic 1						-0.5688***	-0.3599*	0.3291		
						(0.1473)	(0.2046)	(0.7092)		
Demographic 2						-0.0225	-0.0641	-0.7626		
						(0.1609)	(0.2095)	(0.7932)		
Advanced Math							0.5402***	0.3977		
							(0.1471)	(0.6444)		
Committee Score							0.1927***	-0.6714***		
							(0.0410)	(0.1978)		
Missing Committee Score							0.1774			
							(0.1951)			
Year Transition/Matriculate More		-1.0924***	43.58%		96.46%	-0.0704	-0.0765	2.9420***		
		(0.2510)	(37.09%, 50.29%)		(13.07%, 99.98%)	(0.1661)	(0.1767)	(0.6724)		
Program Rank						-0.0307***	-0.0305***			
						(0.0019)	(0.0019)			
$\tilde{m}_2$	0.8882***									
	(0.0488)									
$\overline{m_{2A}}$				4.3874						
				(5.6885)						
$\sqrt{\text{MSE}}$	0.0010***									
	(0.0002)									
Beta									0.8225***	
									(0.0434)	
Sigma										0.2234***
										(0.0318)
Obs | Param | LL | AIC	1000	26	351.0	-650.0						

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 3: Simulated Unrestricted Waitlist-Hybrid Estimates

(a) Optimal Committee

	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success 1		Success 3		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-0.0249	-11.5885***	-2.2558***	-11.9709***	-1.6176***	-13.2754***	-1.6998***	-8.1374***	-1.1678***	-9.4807***	-1.3248***	-5.9143	-0.6836
	(0.0516)	(1.0232)	(0.1454)	(2.5515)	(0.3154)	(2.4119)	(0.2845)	(1.1381)	(0.1344)	(1.3435)	(0.1562)	(6.4966)	(0.7703)
GRE Quantitative Percentile		0.1236***	0.0241***	0.1097***	0.0148***	0.1387***	0.0178***	0.1079***	0.0155***	0.1150***	0.0161***	0.0842	0.0097
		(0.0118)	(0.0018)	(0.0271)	(0.0034)	(0.0265)	(0.0032)	(0.0133)	(0.0015)	(0.0145)	(0.0016)	(0.0645)	(0.0075)
Gender		0.2266	0.0441	0.0234	0.0032	0.3040	0.0389	0.1399	0.0201	0.2078	0.0290	0.4647	0.0537
		(0.1464)	(0.0283)	(0.2899)	(0.0392)	(0.3288)	(0.0421)	(0.1739)	(0.0249)	(0.1766)	(0.0246)	(0.9007)	(0.1052)
Demographic 1		-0.3014	-0.0587	0.5534	0.0748	-0.8832*	-0.1131*	-0.2533	-0.0364	-0.3380	-0.0472	-0.2627	-0.0304
		(0.1988)	(0.0386)	(0.4942)	(0.0665)	(0.4623)	(0.0578)	(0.2359)	(0.0338)	(0.2456)	(0.0342)	(0.7742)	(0.0879)
Demographic 2		0.1538	0.0299	0.5137	0.0694	-0.0523	-0.0067	-0.1179	-0.0169	-0.1354	-0.0189	-0.3955	-0.0457
		(0.2215)	(0.0431)	(0.5205)	(0.0700)	(0.4528)	(0.0580)	(0.2581)	(0.0371)	(0.2652)	(0.0371)	(1.2095)	(0.1391)
Advanced Math				0.5067*	0.0685*	0.8622***	0.1104***			0.6827***	0.0954***	1.1444	0.1323
				(0.2907)	(0.0390)	(0.3306)	(0.0421)			(0.1770)	(0.0237)	(0.9852)	(0.1080)
Committee Score				0.1061	0.0143	0.0378	0.0048			0.0467	0.0065	-0.9116**	-0.1054***
				(0.0689)	(0.0093)	(0.0672)	(0.0086)			(0.0672)	(0.0094)	(0.3783)	(0.0317)
Missing Committee Score										0.2177	0.0304		
										(0.1896)	(0.0264)		
Year Transition/Matriculate More		0.9840***	0.1915***					-0.0587	-0.0084	-0.0466	-0.0065	3.3941***	0.3923***
		(0.1472)	(0.0263)					(0.1690)	(0.0242)	(0.1740)	(0.0243)	(0.7689)	(0.0698)
Program Rank								-0.0310***	-0.0045***	-0.0309***	-0.0043***		
								(0.0019)	(0.0002)	(0.0019)	(0.0002)		
$\dot{m}_2$	1.0896***			-1.5992**	-0.2161**								
	(0.0851)			(0.7875)	(0.1049)								
$\overline{m_{2A}}$						-2.0178***	-0.2584***						
						(0.7700)	(0.0964)						
$\sqrt{\text{MSE}}$	0.0015***												
	(0.0005)												
Obs | Param | LL | AIC	1000	42	191.4	-298.8									

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success 1		Success 3		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-0.0496*	-11.2295***	-2.1801***	-6.7113***	-1.0637***	-7.1639***	-0.8002***	-8.2825***	-1.1913***	-9.7587***	-1.3686***	-1.5054	-0.2179
	(0.0259)	(1.0144)	(0.1474)	(2.0007)	(0.3043)	(2.5070)	(0.2796)	(1.1392)	(0.1341)	(1.3345)	(0.1538)	(3.8095)	(0.5534)
GRE Quantitative Percentile		0.1226***	0.0238***	0.0387*	0.0061*	0.0717***	0.0080**	0.1097***	0.0158***	0.1175***	0.0165***	0.0362	0.0052
		(0.0117)	(0.0018)	(0.0215)	(0.0034)	(0.0278)	(0.0031)	(0.0133)	(0.0015)	(0.0145)	(0.0016)	(0.0435)	(0.0063)
Gender		0.2575*	0.0500*	-0.2767	-0.0439	0.4102	0.0458	0.1355	0.0195	0.1984	0.0278	-0.3519	-0.0509
		(0.1469)	(0.0283)	(0.2746)	(0.0431)	(0.3417)	(0.0380)	(0.1741)	(0.0250)	(0.1766)	(0.0247)	(0.5435)	(0.0788)
Demographic 1		-0.8293***	-0.1610***	0.2577	0.0408	-1.1548**	-0.1290**	-0.2776	-0.0399	-0.3897	-0.0547	0.3291	0.0476
		(0.1966)	(0.0371)	(0.3774)	(0.0594)	(0.5110)	(0.0563)	(0.2344)	(0.0337)	(0.2462)	(0.0343)	(0.7092)	(0.1031)
Demographic 2		0.0797	0.0155	0.5369	0.0851	-0.3696	-0.0413	-0.1809	-0.0260	-0.2068	-0.0290	-0.7626	-0.1104
		(0.2135)	(0.0414)	(0.3886)	(0.0605)	(0.4652)	(0.0521)	(0.2566)	(0.0369)	(0.2636)	(0.0370)	(0.7932)	(0.1090)
Advanced Math				0.2363	0.0375	0.4866	0.0544			0.6387***	0.0896***	0.3977	0.0576
				(0.2639)	(0.0417)	(0.3566)	(0.0399)			(0.1763)	(0.0238)	(0.6444)	(0.0922)
Committee Score				0.3737***	0.0592***	0.0653	0.0073			0.0626	0.0088	-0.6714***	-0.0972***
				(0.0674)	(0.0096)	(0.0837)	(0.0093)			(0.0657)	(0.0092)	(0.1978)	(0.0217)
Missing Committee Score										0.2535	0.0356		
										(0.1896)	(0.0264)		
Year Transition/Matriculate More		1.1108***	0.2156***					-0.0767	-0.0110	-0.0473	-0.0066	2.9420***	0.4259***
		(0.1479)	(0.0257)					(0.1688)	(0.0243)	(0.1757)	(0.0247)	(0.6724)	(0.0579)
Program Rank								-0.0304***	-0.0044***	-0.0303***	-0.0043***		
								(0.0019)	(0.0002)	(0.0019)	(0.0002)		
$\dot{m}_2$	0.8882***			-0.5496	-0.0871								
	(0.0488)			(0.5706)	(0.0900)								
$\overline{m_{2A}}$						-2.9219***	-0.3264***						
						(0.9936)	(0.1102)						
$\sqrt{\text{MSE}}$	0.0010***												
	(0.0002)												
Obs | Param | LL | AIC	1000	42	374.8	-665.5									

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 4: Simulated Restricted Waitlist-Hybrid Estimates

(a) Optimal Committee

	$\overline{m_{2A}}$	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Year Transition/Matriculate Fewer | Intercept	-0.0249	0.6925***	66.65%	0.4942	70.64%	-8.0628***	-9.5820***	-5.9143		
	(0.0516)	(0.1818)	(58.33%, 74.05%)	(0.3513)	(57.55%, 81.02%)	(1.1622)	(1.3341)	(6.4966)		
GRE Quantitative Percentile						0.1062***	0.1154***	0.0842		
						(0.0134)	(0.0146)	(0.0645)		
Gender						0.1999**	0.1696	0.4647		
						(0.1018)	(0.1358)	(0.9007)		
Demographic 1						-0.2482*	-0.3098	-0.2627		
						(0.1353)	(0.1912)	(0.7742)		
Demographic 2						0.0407	-0.0318	-0.3955		
						(0.1511)	(0.1962)	(1.2095)		
Advanced Math							0.6616***	1.1444		
							(0.1424)	(0.9852)		
Committee Score							0.0560	-0.9117**		
							(0.0425)	(0.3783)		
Missing Committee Score							0.2167			
							(0.1838)			
Year Transition/Matriculate More		-0.8491***	46.09%		88.08%	-0.0604	-0.0289	3.3941***		
		(0.2179)	(39.64%, 52.68%)		(65.79%, 96.60%)	(0.1667)	(0.1727)	(0.7689)		
Program Rank						-0.0310***	-0.0310***			
						(0.0019)	(0.0019)			
$\hat{m}_2$	1.0896***									
	(0.0851)									
$\overline{m_{2A}}$				2.3665*						
				(1.2823)						
$\sqrt{\text{MSE}}$	0.0015***									
	(0.0005)									
Beta									0.9490***	
									(0.0391)	
Sigma										0.1926***
										(0.0246)
Obs | Param | LL | AIC	1000	26	186.8	-321.6						

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	$\overline{m_{2A}}$	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Year Transition/Matriculate Fewer | Intercept	-0.0496*	0.8342***	69.72%	0.8216	80.71%	-8.6331***	-8.3628***	-1.5054		
	(0.0259)	(0.2171)	(60.07%, 77.90%)	(0.7603)	(61.39%, 91.67%)	(1.1831)	(1.4735)	(3.8095)		
GRE Quantitative Percentile						0.1147***	0.0947***	0.0362		
						(0.0137)	(0.0156)	(0.0435)		
Gender						0.2301**	0.1269	-0.3519		
						(0.1099)	(0.1412)	(0.5435)		
Demographic 1						-0.5688***	-0.3599*	0.3291		
						(0.1473)	(0.2046)	(0.7092)		
Demographic 2						-0.0225	-0.0641	-0.7626		
						(0.1609)	(0.2095)	(0.7932)		
Advanced Math							0.5402***	0.3977		
							(0.1471)	(0.6444)		
Committee Score							0.1927***	-0.6714***		
							(0.0410)	(0.1978)		
Missing Committee Score							0.1774			
							(0.1951)			
Year Transition/Matriculate More		-1.0924***	43.58%		96.46%	-0.0704	-0.0765	2.9420***		
		(0.2510)	(37.09%, 50.29%)		(13.07%, 99.98%)	(0.1661)	(0.1767)	(0.6724)		
Program Rank						-0.0307***	-0.0305***			
						(0.0019)	(0.0019)			
$\hat{m}_2$	0.8882***									
	(0.0488)									
$\overline{m_{2A}}$				4.3874						
				(5.6885)						
$\sqrt{\text{MSE}}$	0.0010***									
	(0.0002)									
Beta									0.8225***	
									(0.0434)	
Sigma										0.2234***
										(0.0318)
Obs | Param | LL | AIC	1000	26	351.0	-650.0						

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

A.3 Symmetric Waitlist Models

In Appendix A.1, I present waitlist models in which success probabilities equal, the committee is more likely to give Round 2 acceptances to likely matriculants. Intuitively, this primary channel holds because likely matriculants decrease the expected number of spots to fill in Round 3, letting the committee be more selective. But there is also a secondary channel: if the committee accepts likely matriculants in Round 2, then the unlikely matriculants it waitlists end up in the Round 3 pool. As such, while there are fewer expected spots to fill, it is easier to fill those spots. For tractability, I do not estimate this secondary channel in the main specification. But tractability aside, there are two economic reasons why the primary channel is dominant. First, matriculation probabilities decline between Rounds 2 and 3, and for likely matriculants, they have more room to fall. Second, matriculation probabilities can change unpredictably between those rounds.

Still, I present symmetric waitlist models that account for both the primary and secondary channels. There is are two additional states: $\overline{m_3}$, which is a given year's average waitlisted applicant matriculation probability, and m_3, which is the *ex-ante* probability that an applicant will matriculate if waitlisted. By *ex-ante*, I mean that m_3 depends only on information in the application files and not events that happen between Rounds 2 and 3. Therefore, the committee knows m_3 in Round 2. As with $\overline{m_{2A}}$, the Round 3 cutoff probability is increasing in $\overline{m_3}$. In addition, $\overline{m_3}$ is increasing in m_3, analogously to how $\overline{m_{2A}}$ is increasing in $\ddot{m}_2$. Whether the committee will be more or less likely to accept likely matriculants in Round 2 is theoretically ambiguous and depends on the parameter values. The equations are as follows, starting with Round 3:

$$v_3(x,z,t,\overline{k},\overline{m_{2A}},\overline{m_3},\varepsilon_3) = \max_{d_3\in\{A,R\}} u_3(x,z,t,\overline{k},\overline{m_{2A}},\overline{m_3},d_3)+\varepsilon_3(d_3), \text{ such that:} \tag{10}$$

$$u_3(x,z,t,\overline{k},\overline{m_{2A}},\overline{m_3},d_3) = P(y=S|x,z,t,\overline{k})\mathbb{1}(d_3=A)+P(y_3^*=S|\overline{m_{2A}},\overline{m_3})\mathbb{1}(d_3=R).$$

The Round 2 value function is:

$$v_2(x,z,t,\overline{k},\ddot{m}_2,m_3,\varepsilon_2) = \max_{d_2\in\{A,W\}} u_2(x,z,t,\overline{k},\ddot{m}_2,m_3,d_2)+\varepsilon_2(d_2), \text{ such that:} \tag{11}$$

$$u_2(x,z,\ddot{m}_2,m_3,d_2) = P(y=S|x,z,t,\overline{k})\mathbb{1}(d_2=A)+\beta\mathbb{E}[v_3(x,z,t,\overline{k},\overline{m_{2A}},\overline{m_3},\varepsilon_3)|x,z,t,\overline{k},\ddot{m}_2,m_3]\mathbb{1}(d_2=W), \text{ such that:}$$

$$\mathbb{E}[v_3(x,z,t,\overline{k},\overline{m_{2A}},\widetilde{\overline{m_3}},\varepsilon_3)|x,z,t,\overline{k},\ddot{m}_2,m_3] = \int_{\widetilde{\overline{m_{2A}}}}\int_{\widetilde{\overline{m_3}}}\sigma\log\left(\sum_{\widetilde{d}_3\in\{A,R\}} e^{u_3(x,z,t,\overline{k},\widetilde{\overline{m_{2A}}},\widetilde{\overline{m_3}},\widetilde{d}_3)/\sigma}\right)dF_{\overline{m_3}|m_3}(\widetilde{\overline{m_3}}|m_3)\,dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2).$$

Finally, the Round 1 value function is:

$$v_1(x,t,\overline{k}) = \max_{d_1\in\{T,R\}} u_1(x,t,\overline{k},d_1)+\varepsilon_1(d_1), \text{ such that:} \tag{12}$$

$$u_1(x,t,\overline{k},d_1) = \mathbb{E}[v_2(x,z,t,\overline{k},\ddot{m}_2,m_3,\varepsilon_2)|x,t,\overline{k}]\mathbb{1}(d_1=T)+L(y_1^*=S|t)\mathbb{1}(d_1=R), \text{ such that:}$$

$$\mathbb{E}[v_2(x,z,t,\overline{k},\ddot{m}_2,m_3,\varepsilon_2)|x,t\overline{k}] = \int_{\widetilde{z}}\sigma\log\left(\sum_{\widetilde{d}_2\in\{A,R\}} e^{u_2(x,\widetilde{z},t,\overline{k},\ddot{m}_2(x,\widetilde{z},t),m_3(x,\widetilde{z},t),\widetilde{d}_2)/\sigma}\right)dF_{z|x,t}(\widetilde{z}|x,t).$$

In the symmetric waitlist-hybrid model, the Round 1 utility function is instead:

$$u_1(x,t,\bar{k},d_1) = P(y=S|x,t,\bar{k})\mathbb{1}(d_1=T) + P(y_1^*=S|t)\mathbb{1}(d_1=R). \tag{13}$$

Expressed as simply as possible, the restricted CCPs are:

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states},d_r)/\sigma)}{\sum_{\widetilde{d}_r\in D_r}\exp(u_r(\text{states},\widetilde{d}_r)/\sigma)}. \tag{14}$$

The unrestricted CCPs, success probabilities, and cutoff level/probability are $P_{d_1} = P\big(d_1=T\big|x,t;\theta_T\big)$, $P_{d_2} = P\big(d_2=A\big|x,z,\ddot{m}_2,m_3;\theta_{A_2}\big)$, $P_{d_3} = P\big(d_3=A\big|x,z,\overline{m_{2A}},\overline{m_3};\theta_{A_3}\big)$, $P_y = P\big(y=S\big|x,z,t,\bar{k};\theta_S\big)$, $L_{y_1^*} = L\big(y_1^*=S\big|t;\theta_{S_1^*}\big)$, and $P_{y_3^*} = P\big(y_3^*=S\big|\overline{m_{2A}},\overline{m_3};\theta_{S_3^*}\big)$. They have the same specifications as the probabilities and levels earlier in the paper; see Equations (11)–(13) in Section 4.1.2.

Let applicants 1 to N_W be waitlisted, N_W+1 to N_T be accepted in Round 2, and N_T+1 to N be rejected in Round 1. Then the unrestricted symmetric waitlist-dynamic likelihood is:

$$\begin{aligned}\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} f\big(z_n^c|z_n^{\neg c},x_n,t_n;\theta_{F_z}\big) f\big(\overline{m_{2A,n}}|\ddot{m}_{2,n};\theta_{F_{m_2}}\big) f\big(\overline{m_{3,n}}|m_{3,n};\theta_{F_{m_3}}\big)\\
P_{d_{1,n}}(\theta_T)P_{d_{2,n}}(\theta_{A_2})^{\mathbb{1}(d_{2,n}=A)}(1-P_{d_{2,n}}(\theta_{A_2}))^{\mathbb{1}(d_{2,n}=W)}\prod_{n=1}^{N_W}P_{d_{3,n}}(\theta_{A_3})^{\mathbb{1}(d_{3,n}=A)}(1-P_{d_{3,n}}(\theta_{A_3}))^{\mathbb{1}(d_{3,n}=R)}\\
\prod_{n=N_T+1}^{N}\big(1-P_{d_{1,n}}(\theta_T)\big)\prod_{n=1}^{N} f\big(z_n^{\neg c}|x_n,t_n;\theta_{F_z}\big)P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}.\end{aligned} \tag{15}$$

In addition, m_2 and m_3 are parameterized as follows: $m_2 = P_{m_2} = P\big(m_2=M\big|x,z,t;\theta_{M_2}\big)$ and $m_3 = P_{m_3} = P\big(m_3=M\big|x,z,t;\theta_{M_3}\big)$. To estimate θ_{M_2} and θ_{M_3}, I would use the first-stage likelihoods: $\mathcal{L}_{N_T-N_W}^{M_2}(\theta_{M_2}) = \prod_{n=N_W+1}^{N_T}P_{m_{2,n}}(\theta_{M_2})^{\mathbb{1}(m_{2,n}=M)}(1-P_{m_{2,n}}(\theta_{M_2}))^{\mathbb{1}(m_{2,n}=D)}$ and $\mathcal{L}_{N_W}^{M_3}(\theta_{M_3}) = \prod_{n=1}^{N_W}P_{m_{3,n}}(\theta_{M_3})^{\mathbb{1}(d_{3,n}=A,m_{3,n}=M)}(1-P_{m_{3,n}}(\theta_{M_3}))^{\mathbb{1}(d_{3,n}=A,m_{3,n}=D)}$. The samples for estimating m_2 and m_3 are applicants accepted in Rounds 2 and 3 respectively. Since m_2 and m_3 are generated regressors, the standard errors are those from Murphy and Topel (1985) and Hardin (2002).

Finally, in Appendix B, I present two different infinite-horizon multiyear models. A limitation of the waitlist models is that while they give early acceptances to applicants with higher matriculation probabilities all else equal, matriculation probabilities do not affect the probability of a late acceptance. For this reason, the first multiyear model is constructed so that an applicant's overall acceptance probability is increasing in their matriculation probability, the logic being that more matriculants this year will give the committee the option to be more selective next year. Meanwhile, the second multiyear model rewards more diverse applicants in a scenario where diversity increases an applicant's success probability. For example, by accepting more theory applicants this year, the committee may have to worry less next year about having too many applied students relative to applied faculty advisors. But this is just one example of such a scenario.

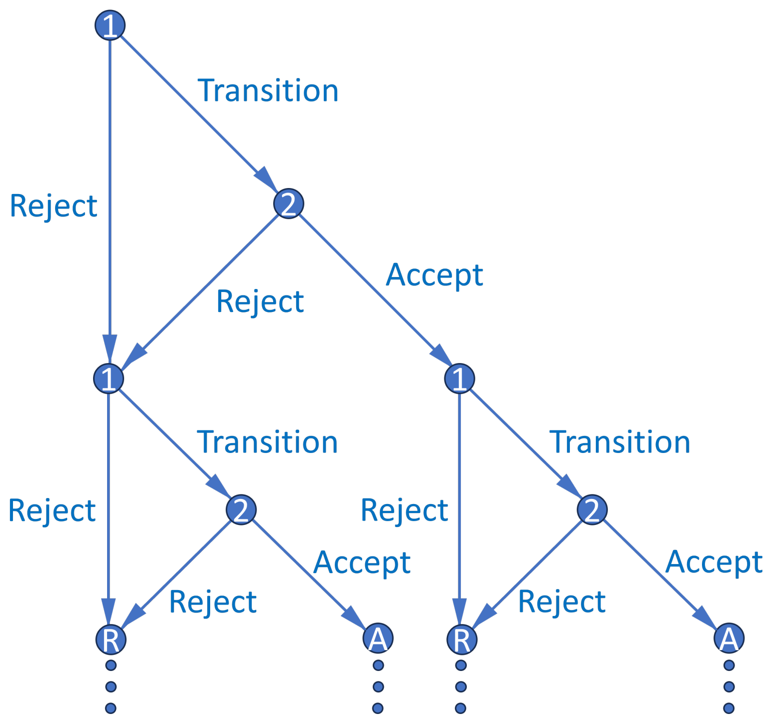


Figure 3: Multiyear Decision Graph

B Multiyear Models

There are two different multiyear models. In the first model, the memory variable is the size of the previous year's cohort. The committee accepts applicants more likely to matriculate because it wants to ensure that next year's memory variable, which is this year's cohort size, is sufficiently large that it can avoid accepting too many relatively weak applicants next year. In the second model, the memory is a diversity variable that increases an applicant's success probability; for example, the previous year's cohort's main area of interest. Applicants with a diverse area of interest may be more likely to succeed because their advisor, with fewer other students to advise, can give them more individualized attention. The committee will therefore accept applicants with a less common area of interest to help balance out the program for next year's class.

B.1 Matriculation Model

In the waitlist models in Appendix A, the committee benefits from giving early acceptances to applicants who are more likely to matriculate. Doing so helps prevent the program from losing out on good applicants who would likely have matriculated if given an early offer but matriculate elsewhere due to events that occurred while on the waitlist. However, the waitlist models do not imply that the committee should also give more late acceptances to likely matriculants. To model that phenomenon, a multiyear model is necessary, as there must be a period after the final round in any given year for dynamics to matter in that round. Figure 3 is a decision graph of the simplest possible multiyear model, in which each year has only two rounds, as in the dynamic model. This model is infinite horizon, and future years are discounted with factor $\beta \in (0,1)$. To avoid the considerations of two-stage estimation, I suppose for simplicity that matriculation probabilities m come from the committee's assessment of each applicant in Round 2. I also suppose that success probabilities depend only on applicant characteristics and not program rank.

In each year's Round 2 value function, I specify the cutoff probability as $P(y_2^* = S|c_{-1})$ instead of $P(y_2^* = S|t)$, where c_{-1} is the previous year's cohort size. The cutoff probability increases in c_{-1} because the committee must keep the total number of students in the program across years relatively constant. So if c_{-1} is large, the committee must now accept fewer students.

Given that, what information available at the start of Round 2 determines c? If and only if the applicant is accepted, their own matriculation probability m does. Now suppose the committee maximizes discounted "lifetime" cohort success by solving an infinite-horizon dynamic programming problem across years. As in the waitlist models, the optimal dynamic decision may not match the optimal myopic one. Instead, the committee may accept an applicant with a high m even if their expected success probability is less than what the Round 2 cutoff level (which need not be between 0 and 1) would otherwise be. The reason is that accepting a high m applicant increases c's distribution, which increases next year's Round 2 cutoff and value.

Intuitively, even if the committee can accept more applicants next year to make up for a smaller cohort this year, they will do so out of necessity, for example, to avoid future TA shortages. This situation will be undesirable provided next year's marginal applicant is of lower quality than this year's above-marginal applicant. Also, as in the waitlist models, even if the data show that the committee does not think dynamically, if c_{-1} affects the Round 2 cutoff, perhaps it should. Finally, estimating this model is more computationally challenging since it does not have a closed-form solution. Nested fixed point algorithms suffer from the curse of dimensionality, and the number of state variables in application files, in particular those from textual documents, is large. But principal component analysis can reduce the size of the state space.

I now present the value functions, starting with each year's Round 2 value:

$$v_2(x,z,t,m,c_{-1},\varepsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x,z,t,m,c_{-1},d_2) + \varepsilon_2(d_2), \text{ such that:} \tag{16}$$

$$\begin{aligned} u_2(x,z,t,m,c_{-1},d_2) = &\{P(y=S|x,z,t) + \beta\mathbb{E}[v_1(x',t',c,\varepsilon_1{}')|m]\}\mathbb{1}(d_2 = A) \\ &+ \{L(y_2^* = S|c_{-1}) + \beta\mathbb{E}[v_1(x',t',c,\varepsilon_1{}')]\}\mathbb{1}(d_2 = R), \text{ such that:} \end{aligned}$$

$$\mathbb{E}[v_1(x',t',c,\varepsilon_1{}')|(m)] = \int_c \int_{t'} \int_{x'} \sigma \log\left(\sum_{d_1{}' \in \{T,R\}} e^{u_1(x',t',c,d_1{}')/\sigma}\right) dF_x(x') dF_t(t') dF_{c|(m)}(c|(m))$$

$$= \int_c \int_{t_1{}'} \cdots \int_{t_{J_t}{}'} \int_{x_1{}'} \cdots \int_{x_{J_x}{}'} \Big[\cdot\Big] dF_{x,1}(x_1{}'|x_2{}',\ldots,x_{J_x}{}') \cdots dF_{x,J_x}(x_{J_x}{}') dF_{t,1}(t_1{}'|t_2{}',\ldots,t_{J_t}{}') \cdots dF_{t,J_t}(t_{J_t}{}') dF_{c|(m)}(c|(m)).$$

As such, Round 2's conditional choice probabilities are:

$$P(d_2|x,z,t,m,c_{-1}) = \frac{\exp(u_2(x,z,t,m,c_{-1},d_2)/\sigma)}{\sum_{\widetilde{d_2}\in\{A,R\}}\exp(u_2(x,z,t,m,c_{-1},\widetilde{d_2})/\sigma)}. \tag{17}$$

Each year's Round 1 value function is:

$$v_1(x,t,c_{-1},\varepsilon_1) = \max_{d_1\in\{T,R\}} u_1(x,t,c_{-1},d_1)+\varepsilon_1(d_1), \text{ such that:} \tag{18}$$

$$u_1(x,t,c_{-1},d_1) = \mathbb{E}[v_2(x,z,t,m,c_{-1},\varepsilon_2)|x,t,c_{-1}]\mathbb{1}(d_1=T)+\{L(y_1^*=S|t)+\beta\mathbb{E}[v_1(x',t',c,\varepsilon_1{}')]\}\mathbb{1}(d_1=R), \text{ such that:}$$

$$\begin{aligned}\mathbb{E}[v_2(x,z,t,m,c_{-1},\varepsilon_2)|x,t,c_{-1}] &= \int_{\widetilde{m}}\int_{\widetilde{z}}\sigma\log\left(\sum_{\widetilde{d_2}\in\{A,R\}} e^{u_2(x,\widetilde{z},t,\widetilde{m},c_{-1},\widetilde{d_2})/\sigma}\right)dF_{z|x,t}(\widetilde{z}|x,t)dF_{m|x,t}(\widetilde{m}|x,t)\\ &= \int_{\widetilde{m}}\int_{\widetilde{z_1}}\cdots\int_{\widetilde{z_{J_z}}}\Big[\cdot\Big]dF_{z|x,t,1}(\widetilde{z_1}|\widetilde{z_2},\ldots,\widetilde{z_{J_z}},x,t)\cdots dF_{z|x,t,J_z}(\widetilde{z_{J_z}}|x,t)dF_{m|x,t}(\widetilde{m}|x,t).\end{aligned}$$

And as such, Round 1's conditional choice probabilities (CCPs) are:

$$P(d_1|x,t,c_{-1}) = \frac{\exp(u_1(x,t,c_{-1},d_1)/\sigma)}{\sum_{\widetilde{d_1}\in\{T,R\}}\exp(u_1(x,t,c_{-1},\widetilde{d_1})/\sigma)}. \tag{19}$$

The unrestricted CCPs, success probabilities, and cutoff levels are $P_{d_1} = P\big(d_1=T\big|x,t,c_{-1};\theta_T\big)$, $P_{d_2} = P\big(d_2=A\big|x,z,m,c_{-1};\theta_A\big)$, $P_y = P\big(y=S\big|x,z,t;\theta_S\big)$, $L_{y_1^*} = L\big(y_1^*=S\big|t;\theta_{S_1^*}\big)$, and $L_{y_2^*} = L\big(y_2^*=S\big|c_{-1};\theta_{S_2^*}\big)$, which aside from the cutoffs have the usual logistic specification. In this model, the cutoffs are not in a function that restricts their range. Given that, the unrestricted likelihood is:

$$\begin{aligned}\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_A} f\big(c_n|m_n;\theta_{F_c}\big)\prod_{n=1}^{N_T} f\big(z_n^c|z_n^{\neg c},x_n,t_n;\theta_{F_z}\big)f\big(m_n|x_n,t_n;\theta_{F_m}\big)P_{d_{1,n}}(\theta_T)P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)}(1-P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}\\ \prod_{n=N_T+1}^{N}\big(1-P_{d_{1,n}}(\theta_T)\big)\prod_{n=1}^{N} f\big(x_n;\theta_{F_x}\big)f\big(t_n;\theta_{F_t}\big)f\big(z_n^{\neg c}|x_n,t_n;\theta_{F_z}\big)P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}.\end{aligned} \tag{20}$$

As with the dynamic models in the paper, I use (θ_F^U,θ_S^U) as an initial guess of (θ_F^R,θ_S^R). To show that the dynamics of this model work as intended, I prove the following theorem:

Theorem 2. *Success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.*

Proof. Suppose there are two arbitrary applicants in year t indexed by H and L, such that $P(y=S|x^H,z^H,t) = P(y=S|x^L,z^L,t)$, but $m^H > m^L$. That is, H has the higher matriculation probability. I will show that $P(d_2=A|x^H,z^H,t,m^H,c_{-1}) > P(d_2=A|x^L,z^L,t,m^L,c_{-1})$.

Define for each round the choice-specific payoffs for acceptance (A) and rejection (R):

$$\pi_2^A \equiv P(y=S|x,z,t)+\beta\mathbb{E}[v_1(x',t',c,\varepsilon_1')|m], \quad \pi_2^R \equiv L(y_2^*=S|c_{-1})+\beta\mathbb{E}[v_1(x',t',c,\varepsilon_1')],$$

$$\pi_1^A \equiv \mathbb{E}[v_2(x',z',t',m',c,\varepsilon_2')|x',t',c], \quad \pi_1^R \equiv L(y_1^{*\prime}=S|t') + \mathbb{E}[v_1(x'',t'',c',\varepsilon_1'')]. \tag{21}$$

It is the case that:

$$P(d_2=A|x,z,t,m,c_{-1}) = \frac{\exp(\pi_2^A/\sigma)}{\exp(\pi_2^A/\sigma)+\exp(\pi_2^R/\sigma)}, \text{ where:}$$

$$\mathbb{E}[v_1(x',t',c,\varepsilon_1')|m] = \int_c \int_{t'} \int_{x'} \sigma \log\left(e^{\pi_1^A/\sigma}+e^{\pi_1^R/\sigma}\right) dF_x(x')\, dF_t(t')\, dF_{c|m}(c|m), \text{ where:}$$

$$\mathbb{E}[v_2(x',z',t',m',c,\varepsilon_2')|x',t',c] = \int_{\widetilde{m'}} \int_{\widetilde{z'}} \sigma \log\left(e^{\pi_2^{A\prime}/\sigma}+e^{\pi_2^{R\prime}/\sigma}\right) dF_{z|x,t}(\widetilde{z'}|x',t') dF_{m|x,t}(\widetilde{m'}|x',t'), \tag{22}$$

where $\pi_2^{A\prime}$ and $\pi_2^{R\prime}$ are defined analogously to Equation (21) at the next year's states.

Per-period utilities are bounded and $\beta \in (0,1)$, so the value functions exist and are unique. The only place m appears in $P(d_2=A|x,z,t,m,c_{-1})$ is in the distribution of c conditional on m. I parameterize $c|m \sim \mathcal{N}(m\beta_m, \sigma_m^2)$, where by assumption $\beta_m > 0$. β_m is positive because higher matriculation probabilities lead to a greater expected cohort size. Multiple years of data are necessary to identify β_m, since cohort sizes vary only across years. Then:

$$F_{c|m}(c|m^H;\beta_m,\sigma_m^2) = \Phi\left(\frac{c-m^H\beta_m}{\sigma_m}\right) < \Phi\left(\frac{c-m^L\beta_m}{\sigma_m}\right) = F_{c|m}(c|m^L;\beta_m,\sigma_m^2). \tag{23}$$

Therefore, $c|m^H$ first-order stochastically dominates $c|m^L$. To apply this stochastic dominance, I must show that $\mathbb{E}[v_1(x',t',c,\varepsilon_1')|m]$'s integrand consisting of all but the outermost integral is strictly increasing in c. But the only place c appears there is in $\mathbb{E}[v_2(x',z',t',m',c,\varepsilon_2')|x',t',c]$, and the only place c appears in that expectation is in $L(y_2^{*\prime}=S|c;\theta_{S_2^*}) = f_{S_2^*}(c)\theta_{S_2^*}$.

By assumption, $f_{S_2^*}(\cdot)$ is strictly increasing and $\theta_{S_2^*} > 0$. $\theta_{S_2^*,c}$ is positive because a greater cohort size this year means that the committee, needing to keep the total number of students in the program relatively constant across years, can afford to be more selective next year. As such, I have that $P(y_2^{*\prime}=S|c)$, and in turn $\mathbb{E}[v_1(x',t',c,\varepsilon_1')|m]$'s integrand, are strictly increasing in c. Combining this result with $[c|m^H] \succ_{FOSD} [c|m^L]$, I obtain the inequality:

$$\begin{aligned}
\mathbb{E}[v_1(x',t',c,\varepsilon_1')|m^H] &= \int_c \int_{t'} \int_{x'} \sigma \log\left(e^{\pi_1^A/\sigma}+e^{\pi_1^R/\sigma}\right) dF_x(x')\, dF_t(t')\, dF_{c|m}(c|m^H) \\
&> \int_c \int_{t'} \int_{x'} \sigma \log\left(e^{\pi_1^A/\sigma}+e^{\pi_1^R/\sigma}\right) dF_x(x')\, dF_t(t')\, dF_{c|m}(c|m^L) \\
&= \mathbb{E}[v_1(x',t',c,\varepsilon_1')|m^L].
\end{aligned} \tag{24}$$

Therefore, by Equation (22), Inequality (24), and $P(y=S|x^H,z^H,t) = P(y=S|x^L,z^L,t)$, I have that $P(d_2=A|x^H,z^H,t,m^H,c_{-1}) > P(d_2=A|x^L,z^L,t,m^L,c_{-1})$. Intuitively, if the committee rejects either applicant, their m does not matter. But since their success probabilities are equal, accepting applicant H yields a higher expected payoff than accepting applicant L.

B.2 Diversity Model

I remove the state variable c_{-1} and add in its place two new states a and $\overline{a}_{-1}$. Let $a = \mathbb{1}(\text{Diverse})$, where a diverse characteristic is one whose probability of appearing among next year's applicants is bounded above by some $p \in (0,1)$. For example, I could have $a = \mathbb{1}(\text{TheoryField})$, where in most years, theory applicants are less common than applied applicants. Also, let $\overline{a}_{-1}$ equal the percentage of last year's acceptance set that is theory. If $|a-\overline{a}_{-1}|$ shrinks, then the applicant's expected success probability decreases, as their future advisor may have less time for them.[5]

As such, the optimal dynamic decision may not match the optimal myopic one. In particular, the committee may accept a diverse applicant it would not have otherwise because doing so increases the success probability of next year's applicant pool. Accepting a diverse applicant increases the distribution of $\overline{a}$ conditional on a, which because diverse applicants are less common increases the distribution of $|a'-\overline{a}|$, which increases next year's expected success probability, which increases next year's expected value. That said, I can also estimate a myopic model with memory variable $\overline{a}_{-1}$. The committee will still prioritize diverse applicants because diversity increases applicants' success probabilities, but there will be no forward-looking dynamic effect.

I now present the value functions, starting with each year's Round 2 value:

$$v_2(x,z,t,a,\overline{a}_{-1},\varepsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x,z,t,a,\overline{a}_{-1},d_2) + \varepsilon_2(d_2), \text{ such that:} \tag{25}$$

$$\begin{aligned} u_2(x,z,t,a,\overline{a}_{-1},d_2) = &\{P(y=S|x,z,t,|a-\overline{a}_{-1}|) + \beta\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')|a]\}\mathbb{1}(d_2=A) \\ &+ \{L(y_2^*=S|t) + \beta\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')]\}\mathbb{1}(d_2=R), \text{ such that:} \end{aligned}$$

$$\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')|a] = \int_{\widetilde{\overline{a}}}\int_{\widetilde{t'}}\int_{\widetilde{x'}} \sigma \log\left(\sum_{d_1' \in \{T,R\}} e^{u_1(\widetilde{x'},\widetilde{t'},\widetilde{\overline{a}},d_1')/\sigma}\right) dF_{x'}(\widetilde{x'})dF_{t'}(\widetilde{t'})dF_{\overline{a}|a}(\widetilde{\overline{a}}|a) =$$

$$\int_{\widetilde{\overline{a}}}\int_{\widetilde{t_1'}}\cdots\int_{\widetilde{t_{J_t}'}}\int_{\widetilde{x_1'}}\cdots\int_{\widetilde{x_{J_x}'}}\left[\cdot\right] F_{x,1}({x_1}'|{x_2}',\ldots,{x_{J_x}}')\cdots dF_{x,J_x}({x_{J_x}}')dF_{t,1}({t_1}'|{t_2}',\ldots,{t_{J_t}}')\cdots dF_{t,J_t}({t_{J_t}}')dF_{\overline{a}|a}(\widetilde{\overline{a}}|a).$$

As such, Round 2's conditional choice probabilities are:

$$P(d_2|x,z,t,a,\overline{a}_{-1}) = \frac{\exp(u_2(x,z,t,a,\overline{a}_{-1},d_2)/\sigma)}{\sum_{\widetilde{d_2}\in\{A,R\}}\exp(u_2(x,z,t,a,\overline{a}_{-1},\widetilde{d_2})/\sigma)}. \tag{26}$$

Each year's Round 1 value function is:

$$v_1(x,t,\overline{a}_{-1},\varepsilon_1) = \max_{d_1\in\{T,R\}} u_1(x,t,\overline{a}_{-1},d_1) + \varepsilon_1(d_1), \text{ such that:} \tag{27}$$

[5] If the program never gets too imbalanced, this relationship will not be identifiable, in which case estimation will not work for this choice of a. However, there may also be different suitable diversity variables.

$$u_1(x,t,\overline{a}_{-1},d_1) = \mathbb{E}[v_2(x,z,t,a,\overline{a}_{-1},\varepsilon_2)|x,t,\overline{a}_{-1}]\mathbb{1}(d_1 = T)$$
$$+\{L(y_1^* = S|t) + \beta\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')]\}\mathbb{1}(d_1 = R), \text{ such that:}$$

$$\mathbb{E}[v_2(x,z,t,a,\overline{a}_{-1},\varepsilon_2)|x,t,\overline{a}_{-1}] = \int_{\widetilde{a}}\int_{\widetilde{z}} \sigma \log\left(\sum_{\widetilde{d}_2 \in A,R} e^{u_2(x,\widetilde{z},t,\widetilde{a},\overline{a}_{-1},\widetilde{d}_2)/\sigma}\right) dF_{z|x,t}(\widetilde{z}|x,t)dF_a(\widetilde{a})$$
$$= \int_{\widetilde{a}}\int_{\widetilde{z}_1}\cdots\int_{\widetilde{z}_J}\left[\cdot\right] dF_1(\widetilde{z}_1|\widetilde{z}_2,\ldots\widetilde{z}_J,x,t)\cdots dF_J(\widetilde{z}_J|x,t)dF_a(\widetilde{a})$$

$$\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')] = \int_{\widetilde{\overline{a}}}\int_{\widetilde{t'}}\int_{\widetilde{x'}} \sigma\log\left(\sum_{d_1' \in \{T,R\}} e^{u_1(\widetilde{x'},\widetilde{t'},\widetilde{\overline{a}},d_1')/\sigma}\right) dF_{x'}(\widetilde{x'})dF_{t'}(\widetilde{t'})dF_{\overline{a}}(\widetilde{\overline{a}}) =$$
$$\int_{\widetilde{\overline{a}}}\int_{\widetilde{t_1'}}\cdots\int_{\widetilde{t_{J_t}'}}\int_{\widetilde{x_1'}}\cdots\int_{\widetilde{x_{J_x}'}}\left[\cdot\right] F_{x,1}({x_1}'|{x_2}',\ldots,{x_{J_x}}')\cdots dF_{x,J_x}({x_{J_x}}')dF_{t,1}({t_1}'|{t_2}',\ldots,{t_{J_t}}')\cdots dF_{t,J_t}({t_{J_t}}')dF_{\overline{a}}(\widetilde{\overline{a}}).$$

And as such, Round 1's conditional choice probabilities are:

$$P(d_1|x,t,\overline{a}_{-1}) = \frac{\exp(u_1(x,t,\overline{a}_{-1},d_1)/\sigma)}{\sum_{\widetilde{d}_1 \in \{T,R\}} \exp(u_1(x,t,\overline{a}_{-1},\widetilde{d}_1)/\sigma)}. \tag{28}$$

The unrestricted CCPs, success probabilities, and cutoff levels are $P_{d_1} = P\big(d_1 = T\big|x,t,\overline{a}_{-1};\theta_T\big)$, $P_{d_2} = P\big(d_2 = A\big|x,z,t,a,\overline{a}_{-1};\theta_A\big)$, $P_y = P\big(y = S\big|x,z,t,|a-\overline{a}_{-1}|;\theta_S\big)$, $L_{y_1^*} = L\big(y_1^* = S\big|t;\theta_{S_1^*}\big)$, and $L_{y_2^*} = L\big(y_2^* = S\big|t;\theta_{S_2^*}\big)$; all but the cutoffs are logistic. Given that, the unrestricted likelihood is:

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_A} f\big(\overline{a}_n|a_n;\theta_{F_{\overline{a}}}\big)\prod_{n=1}^{N_T} f\big(z_n^c|z_n^{\neg c},x_n,t_n;\theta_{F_z}\big)f\big(a_n;\theta_{F_a}\big)P_{d_{1,n}}(\theta_T)P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)}(1-P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}$$
$$\prod_{n=N_T+1}^{N}\big(1-P_{d_{1,n}}(\theta_T)\big)\prod_{n=1}^{N} f\big(x_n;\theta_{F_x}\big)f\big(t_n;\theta_{F_t}\big)f\big(z_n^{\neg c}|x_n,t_n;\theta_{F_z}\big)P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}. \tag{29}$$

I estimate $f_{\overline{a}|a}(\overline{a}|a)$ on the N_A accepted applicants only (those from N_A+1 to $N_{T,R}$ are rejected in Round 2) because only accepted applicants can influence the acceptance set's area of interest. Also, as with the dynamic models in the paper, I use (θ_F^U,θ_S^U) as an initial guess of (θ_F^R,θ_S^R). To show that the dynamics of this model work as intended, I prove the following theorem:

Theorem 3. *Consider any two applicants, one with $a = 1$ and one with $a = 0$, but with equal success probabilities and therefore different values of x or z. Suppose $P(a' = 1) = p > 0$. Then for all sufficiently small p, the $a = 1$ applicant has a strictly greater Round 2 acceptance probability.*

Proof. Suppose there are two arbitrary applicants in year t, one with $a = 1$ and the other with $a = 0$, but with $P(y = S|x^1,z^1,t,1-\overline{a}_{-1}) = P(y = S|x^0,z^0,t,\overline{a}_{-1})$. Unlike in the previous two proofs, these equal success probabilities imply that unless $\overline{a}_{-1} = \frac{1}{2}$, it must be that $(x^1,z^1) \neq (x^0,z^0)$. I will show that $P(d_2 = A|x^1,z^1,t,1,\overline{a}_{-1}) > P(d_2 = A|x^0,z^0,t,0,\overline{a}_{-1})$.

Define for each round the choice-specific payoffs for acceptance (A) and rejection (R):

$$\pi_2^A \equiv P(y=S|x,z,t,|a-\overline{a}_{-1}|)+\beta\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')|a], \quad \pi_2^R \equiv L(y_2^*=S|t)+\beta\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')],$$

$$\pi_1^A \equiv \mathbb{E}[v_2(x',z',t',a',\overline{a},\varepsilon_2')|x',t',\overline{a}], \quad \pi_1^R \equiv L(y_1^{*\prime}=S|t')+\beta\mathbb{E}[v_1(x'',t'',\overline{a}',\varepsilon_1'')]. \qquad (30)$$

It is the case that:

$$P(d_2=A|x,z,t,a,\overline{a}_{-1}) = \frac{\exp(\pi_2^A/\sigma)}{\exp(\pi_2^A/\sigma)+\exp(\pi_2^R/\sigma)}, \text{ where:}$$

$$\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')|a] = \int_{\widetilde{\overline{a}}}\int_{\widetilde{t'}}\int_{\widetilde{x'}} \sigma\log\left(e^{\pi_1^A/\sigma}+e^{\pi_1^R/\sigma}\right)dF_x(\widetilde{x'})\,dF_t(\widetilde{t'})\,dF_{\overline{a}|a}(\widetilde{\overline{a}}|a), \text{ where:}$$

$$\begin{aligned}\mathbb{E}[v_2(x',z',t',a',\overline{a},\varepsilon_2')|x',t',\overline{a}] = {}& P(a'=1)\int_{\widetilde{z'}}\sigma\log\left(e^{\pi_2^{A\prime}(1)/\sigma}+e^{\pi_2^{R\prime}/\sigma}\right)dF_{z|x,t}(\widetilde{z'}|x',t') \\ &+[1-P(a'=1)]\int_{\widetilde{z'}}\sigma\log\left(e^{\pi_2^{A\prime}(0)/\sigma}+e^{\pi_2^{R\prime}/\sigma}\right)dF_{z|x,t}(\widetilde{z'}|x',t'), \quad (31)\end{aligned}$$

where $\pi_2^{A\prime}(a')$ denotes next year's Round 2 acceptance payoff evaluated at $a' \in \{0,1\}$.

Per-period utilities are bounded and $\beta \in (0,1)$, so the value functions exist and are unique. The only two places a appears in $P(d_2=A|x,z,t,a,\overline{a}_{-1})$ are in the success probabilities, which are assumed to be equal for the $a=1$ and $a=0$ applicants, and in the distribution of $\overline{a}$ conditional on a. I parameterize $\overline{a}|a \sim \mathcal{N}(a\beta_a,\sigma_a^2)$, where by assumption $\beta_a>0$. β_a is positive because $a=1$ applicants are more likely than $a=0$ applicants to be in a year with a high percentage of accepted $a=1$ applicants. As such, multiple years of data are necessary to identify β_a. Then:

$$F_{\overline{a}|a}(\overline{a}|1;\beta_a,\sigma_a^2) = \Phi\left(\frac{\overline{a}-\beta_a}{\sigma_a}\right) < \Phi\left(\frac{\overline{a}}{\sigma_a}\right) = F_{\overline{a}|a}(\overline{a}|0;\beta_a,\sigma_a^2). \qquad (32)$$

Therefore, $\overline{a}|1$ first-order stochastically dominates $\overline{a}|0$. To apply this stochastic dominance, I must show that $\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')|a]$'s integrand consisting of all but the outermost integral is strictly increasing in $\overline{a}$. But the only place $\overline{a}$ appears there is in $\mathbb{E}[v_2(x',z',t',a',\overline{a},\varepsilon_2')|x',t',\overline{a}]$, and the only place $\overline{a}$ appears in that expectation is in $P(y'=S|x',z',t',\overline{a};\theta_S) = \frac{\exp(f_S(x',z',t',|a'-\overline{a}|)\theta_S)}{1+\exp(f_S(x',z',t',|a'-\overline{a}|)\theta_S)}$.

By assumption, $f_S(\cdot)$ is strictly increasing and $\theta_{S,|a-\overline{a}_{-1}|}>0$. $\theta_{S,|a-\overline{a}_{-1}|}$ is positive because a large value of $|a-\overline{a}_{-1}|$ means that applicant a is diverse relative to the previous year's accepted applicants and is therefore more likely to succeed. Recall $P(a'=1)=p$. Taking the limit $p \to 0$:

$$\begin{aligned}&\mathbb{E}[v_2(x',z',t',a',\overline{a},\varepsilon_2')|x',t',\overline{a}] \\ &\to \int_{\widetilde{z'}}\sigma\log(e^{\{P(y'=S|x',z',t',\overline{a})+\beta\mathbb{E}[v_1(x'',t'',\overline{a}',\varepsilon_1'')|0]\}/\sigma}+e^{\{L(y_2^{*\prime}=S|t')+\beta\mathbb{E}[v_1(x'',t'',\overline{a}',\varepsilon_1'')]\}/\sigma})dF_{z|x,t}(\widetilde{z'}|x',t'). \quad (33)\end{aligned}$$

$P(y'=S|x',z',t',\overline{a})$ is strictly increasing in $\overline{a}$, and by the continuity of $E[v_2(x',z',t',a',\overline{a},\varepsilon_2')|x',t',\overline{a}]$ in p, I have that it, and in turn $E[v_1(x',t',\overline{a},\varepsilon_1')|a]$'s integrand, are also strictly increasing in $\overline{a}$ for sufficiently small p. Combining this result with $\overline{a}|1 \succ_{FOSD} \overline{a}|0$, I obtain the inequality:

$$\begin{aligned}
\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')|1] &= \int_{\widetilde{\overline{a}}}\int_{\widetilde{t'}}\int_{\widetilde{x'}} \sigma \log\left(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}\right) dF_x(\widetilde{x'})\, dF_t(\widetilde{t'})\, dF_{\overline{a}|a}(\widetilde{\overline{a}}|1) \\
&> \int_{\widetilde{\overline{a}}}\int_{\widetilde{t'}}\int_{\widetilde{x'}} \sigma \log\left(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}\right) dF_x(\widetilde{x'})\, dF_t(\widetilde{t'})\, dF_{\overline{a}|a}(\widetilde{\overline{a}}|0) \\
&= \mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')|0].
\end{aligned} \tag{34}$$

Therefore, by Equation (31), Inequality (34), and $P(y=S|x^1,z^1,t,1-\overline{a}_{-1}) = P(y=S|x^0,z^0,t,\overline{a}_{-1})$, I have that $P(d_2=A|x^1,z^1,t,1,\overline{a}_{-1}) > P(d_2=A|x^0,z^0,t,0,\overline{a}_{-1})$. Intuitively, if the committee rejects either applicant, their a does not matter. But since their success probabilities are equal, accepting applicant $a=1$ yields a higher expected payoff than accepting $a=0$.

In this proof, I must assume $P(a'=1)$ is sufficiently small, and not just $P(a'=1) < \frac{1}{2}$. The reason is that if $P(a'=1)$ is too large, then $\mathbb{E}[v_2(x',z',t',a',\overline{a},\varepsilon_2')|x',t',\overline{a}]$ may not be strictly increasing in $\overline{a}$, considering $P(y'=S|x',z',t',1-\overline{a})$ is strictly decreasing in $\overline{a}$. This indeterminacy can be explored by calculating $\frac{\partial}{\partial \overline{a}}\mathbb{E}[v_2(x',z',t',a',\overline{a},\varepsilon_2')|x',t',\overline{a}]$. As such, to be sure the committee is more likely to accept the $a=1$ applicant, the committee must know that there will be sufficiently few $a'=1$ applicants. This assumption is strong, but possible if $a'=1$ is sufficiently rare for the committee to know it will happen. But even without forward-looking dynamics, if $\overline{a}_{-1} < \frac{1}{2}$, then the committee will be more likely to accept an $a=1$ applicant than an $a=0$ applicant with identical x and z values. The reason is that $P(y=S|x,z,t,1-\overline{a}_{-1}) > P(y=S|x,z,t,\overline{a}_{-1})$.

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